

Cognitive Integrity Framework: Trust Calculus and Detection Bounds

This section presents the formal foundations of the Cognitive Integrity Framework (CIF). We define the system model (`sec:system-model`), cognitive state representation (`sec:cognitive-state`), integrity properties (`sec:integrity-properties`), trust calculus (`sec:trust-calculus`), and information-theoretic detection bounds (`sec:detection-bounds`).

System Model

Definition (Multiagent Operator)

A *multiagent operator* is a tuple:

$$\mathcal{O} = \langle \mathcal{A}, \mathcal{C}, \mathcal{S}, \mathcal{P}, \Gamma \rangle \quad (1)$$

where components are defined in `tab:operator-components`.

Table 1: Components of the multiagent operator $\mathcal{O}$.

Component	Symbol	Description
Agents	$\mathcal{A} = \{a_1, \dots, a_n\}$	Finite set of n agents
Communication	$\mathcal{C} : \mathcal{A} \times \mathcal{A} \rightarrow \{0, 1\}$	Adjacency matrix encoding permitted channels
Shared State	$\mathcal{S}$	Observable global state
Permissions	$\mathcal{P} : \mathcal{A} \times \text{Actions} \rightarrow \{0, 1\}$	Action authorization mapping
Protocol	Γ	Coordination and communication rules

Cognitive State

Intuitively, an agent's cognitive state captures everything it believes, wants, intends, and remembers at a given moment. This formal representation enables precise reasoning about how attacks manipulate agent reasoning.

Definition (Agent Cognitive State)

Each agent $a_i \in \mathcal{A}$ maintains cognitive state:

$$\sigma_i = \langle \mathcal{B}_i, \mathcal{G}_i, \mathcal{I}_i, \mathcal{H}_i \rangle \tag{2}$$

with components defined in `tab:cognitive-components`.

Table 2: Cognitive state components for agent a_i .

Component	Formal Type	Semantics
Beliefs	$\mathcal{B}_i : \Phi \rightarrow [0, 1]$	Probability distribution over propositions
Goals	$\mathcal{G}_i = \{(g_k, p_k)\}$	Prioritized objectives where $\sum_k p_k = 1$
Intentions	$\mathcal{J}_i = [(a_1, t_1), \dots]$	Committed action sequence with timing
History	$\mathcal{H}_i = [(e_1, t_1), \dots]$	Interaction trace (events, timestamps)

Definition (System State)

The global system state at time t is:

$$S^t = (\sigma_1^t, \dots, \sigma_n^t, s^t, \mathcal{T}^t) \quad (3)$$

where $\mathcal{T}^t$ denotes the trust matrix at time t .

The following transition rules formalize how agent states evolve. Each rule has the form “if preconditions hold (above the line), then this transition occurs (below the line).” Readers may skim the mathematical details on first reading, returning for precision when needed.

Definition (Transition Relation)

State transitions follow the relation $S^t \xrightarrow{\alpha} S^{t+1}$ where $\alpha \in \{\text{RECEIVE}, \text{UPDATE}, \text{ACT}, \text{COMMUNICATE}\}$.

The transition rules are defined as follows:

Rule T-Receive (Message Reception):

$$\frac{m \in \text{channel}(a_j, a_i) \quad \mathcal{F}(m) = \text{ACCEPT}}{(\sigma_i, \text{inbox}_i) \xrightarrow{\text{RECEIVE}} (\sigma_i, \text{inbox}_i \cup \{m\})} \quad (4)$$

Rule T-Reject (Message Rejection):

$$\frac{m \in \text{channel}(a_j, a_i) \quad \mathcal{F}(m) \in \{\text{REJECT}, \text{QUARANTINE}\}}{(\sigma_i, \text{inbox}_i) \xrightarrow{\text{RECEIVE}} (\sigma_i, \text{inbox}_i)} \quad (5)$$

Rule T-Update (Belief Update):

$$\frac{m \in \text{inbox}_i \quad e = \text{extract}(m) \quad s = \text{source}(m)}{\mathcal{B}_i^t \xrightarrow{\text{UPDATE}} \mathcal{B}_i^{t+1} = \text{BayesUpdate}(\mathcal{B}_i^t, e, \mathcal{J}_{i \rightarrow s})} \quad (6)$$

Rule T-Act (Action Execution):

$$\frac{a \in \mathcal{J}_i \quad \mathcal{P}_{\text{eff}}(a_i, a) = 1 \quad \text{precond}(a, \mathcal{S}^t)}{(\sigma_i, \mathcal{S}^t) \xrightarrow{\text{ACT}} (\sigma'_i, \text{effect}(a, \mathcal{S}^t))} \quad (7)$$

Rule T-Communicate (Message Sending):

$$\frac{\mathcal{C}(a_i, a_j) = 1 \quad m = \text{compose}(\sigma_i)}{(\sigma_i, \text{channel}(a_i, a_j)) \xrightarrow{\text{COMM}} (\sigma_i, \text{channel}(a_i, a_j) \cup \{m\})} \quad (8)$$

Definition (Well-Formed Transition Sequence)

A transition sequence $S^0 \xrightarrow{\alpha_1} \dots \xrightarrow{\alpha_k} S^k$ is well-formed iff:

1. **Causality:** $\forall i : S^i$ enables α_{i+1}
2. **Atomicity:** Each α_i is atomic
3. **Fairness:** No agent is starved indefinitely

Theorem (Determinism)

Given state S^t and action α , the resulting state S^{t+1} is uniquely determined.

Proof.

By case analysis on transition rules

`eq:rule-receive`–`eq:rule-comm`. Each rule specifies unique postconditions. □

Integrity Properties

Integrity Properties I

We define four core integrity properties that CIF aims to preserve.

Property (Belief Consistency)

$$\text{Consistent}(\mathcal{B}_i) \iff \nexists \phi, \psi : \mathcal{B}_i(\phi) > \tau \wedge \mathcal{B}_i(\psi) > \tau \wedge (\phi \wedge \psi \vdash \perp) \quad (9)$$

No high-confidence beliefs contradict each other.

Property (Goal Alignment)

$$\text{Aligned}(\mathcal{G}_i) \iff \mathcal{G}_i \subseteq \mathcal{G}_{\text{principal}} \cup \text{Delegate}(\mathcal{G}_{\text{principal}}) \quad (10)$$

All goals derive from the principal or valid delegation chains.

Property (Provenance Verifiability)

$$\text{Verifiable}(\mathcal{B}_i) \iff \forall \phi : \mathcal{B}_i(\phi) > \tau \Rightarrow \exists \pi(\phi) : V(\pi(\phi)) = 1 \quad (11)$$

Every accepted belief has a verifiable provenance chain π .

Property (Action Authorization)

$$\text{Auth}(a_i, \text{act}) \iff \mathcal{P}(a_i, \text{act}) = 1 \vee \exists a_j : \text{Delegate}(a_j, a_i, \text{act}) \quad (12)$$

Actions require direct permission or valid delegation.

Trust Calculus

Motivation: Why Bounded Trust Matters I

Before presenting the formal trust calculus, we motivate its design through concrete scenarios that illustrate why naive trust models fail in multiagent systems.

The Trust Laundering Problem. Consider an adversary with low direct trust who seeks to influence a high-value agent. In a naive trust model, the adversary could:

Motivation: Why Bounded Trust Matters I

1. Establish contact with a moderately trusted intermediary agent
2. Provide accurate information over time to build trust with the intermediary
3. Use the intermediary to relay adversarial content to the target
4. The target accepts the content because it comes from a “trusted” source

Motivation: Why Bounded Trust Matters I

This is *trust laundering*—converting low-trust origin into high-trust delivery through intermediaries. Without bounded delegation, the adversarial content arrives at the target with the intermediary's trust score, not the adversary's.

The Trust Amplification Problem. In peer-to-peer multiagent architectures, agents form trust relationships bidirectionally. Without constraints, circular trust relationships can amplify trust scores:

Motivation: Why Bounded Trust Matters I

$$A \xrightarrow{0.9} B \xrightarrow{0.9} C \xrightarrow{0.9} A$$

If trust flows around this cycle, naive aggregation could yield trust scores exceeding initial values. Our trust algebra prevents this through the δ^d decay bound.

Real-World Delegation Patterns. Modern agentic systems exhibit deep delegation chains. Consider Claude Code processing a user request:

Motivation: Why Bounded Trust Matters I

1. User requests security audit → Orchestrator agent (trust: principal)
2. Orchestrator delegates to Code Analysis agent (depth 1)
3. Code Analysis queries External CVE Database (depth 2)
4. CVE Database returns vulnerability data (depth 3)
5. Code Analysis delegates to Remediation agent (depth 4)
6. Remediation queries StackOverflow for fix patterns (depth 5)

Motivation: Why Bounded Trust Matters I

At depth 5, should the orchestrator trust StackOverflow content with the same confidence as direct user input? Our trust calculus says no: with $\delta = 0.9$, trust at depth 5 is at most $0.9^5 \approx 0.59$ —sufficient for low-stakes decisions but automatically triggering review for high-stakes actions.

Cross-Modality Trust Challenges. When a vision model processes an image and reports “this diagram shows system architecture,” how should a code generation agent weight this claim? Cross-modality trust introduces additional considerations:

Motivation: Why Bounded Trust Matters I

- ▶ **Modality-specific error rates:** Vision models may have different reliability profiles than language models
- ▶ **Adversarial input susceptibility:** Images are particularly vulnerable to adversarial perturbations
- ▶ **Verification difficulty:** Claims about visual content are harder to verify than claims about text or code

Motivation: Why Bounded Trust Matters I

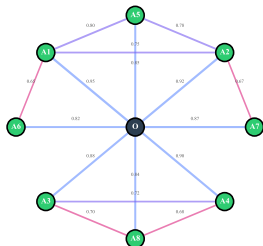
Our framework addresses this through modality-adjusted base trust: $T_{\text{base}}^{\text{vision}} = \eta \cdot T_{\text{base}}^{\text{text}}$ where $\eta < 1$ reflects elevated adversarial risk in visual modalities.

Formal Trust Model I

Formal Trust Model I

$$T = \alpha T_{\text{base}} + \beta T_{\text{rep}} + \gamma T_{\text{ctx}}$$

A. Normal Trust Network



B. Trust Network Under Attack

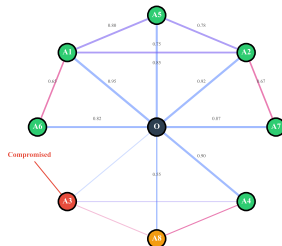


Figure 1: Trust Network Topology: Directed graph showing trust relationships $\mathcal{T}_{i \rightarrow j}$ between agents in hierarchical (left) and peer-to-peer (right) configurations. Edge weights represent trust values in $[0, 1]$; doubled arrows indicate bidirectional trust. Orchestrator α_0 occupies hub position in hierarchical topology.

`fig:trust-network` visualizes the trust relationships in a representative multiagent operator. Edge weights represent trust scores $\mathcal{T}_{i \rightarrow j}$, with thicker edges indicating higher trust. The network topology illustrates how trust propagates through delegation chains and highlights potential attack surfaces for trust manipulation attacks (Ω_4).

Definition (Trust Function)

Trust from agent a_i to agent a_j at time t :

$$\mathcal{T}_{i \rightarrow j}^t = \alpha \cdot T_{\text{base}}(j) + \beta \cdot T_{\text{rep}}^t(j) + \gamma \cdot T_{\text{ctx}}^t(i, j) \quad (13)$$

subject to $\alpha + \beta + \gamma = 1$, with components in
`tab:trust-components`.

Table 3: Trust function components.

Component	Weight	Description
T_{base}	α	Architectural trust (role-based)
T_{rep}	β	Reputation (historical accuracy)
T_{ctx}	γ	Context (task-specific factors)

Remark (Trust as Precision Weighting)

The composite trust formula above is equivalent to precision-weighted message integration from the Free Energy Principle: the composite score $\mathcal{T}_{i \rightarrow j}$ is proportional to the precision ρ_j with which agent i should weight messages from agent j when updating its posterior beliefs. High-trust messages dominate the belief update; low-trust messages contribute marginally. This correspondence provides a neurocomputational grounding for the trust calculus (Part 2, §1.2).

Definition (Trust Delegation)

When agent a_i delegates trust through a_j to a_k :

$$\mathcal{T}_{i \rightarrow k}^{\text{del}} = \min(\mathcal{T}_{i \rightarrow j}, \mathcal{T}_{j \rightarrow k}) \cdot \delta^d \quad (14)$$

where $\delta \in (0, 1)$ is the decay factor and $d \in \mathbb{N}$ is the delegation depth.

The trust algebra provides the mathematical foundation for combining trust scores. The key insight is that trust through intermediaries (delegation, $\otimes$) uses the minimum-then-decay rule, while trust from multiple sources (aggregation, $\oplus$) uses the maximum. This prevents both trust laundering and artificial inflation.

Definition (Trust Algebra)

The trust algebra $(\mathcal{T}, \otimes, \oplus, 0, 1)$ comprises:

- ▶ **Domain:** $\mathcal{T} = [0, 1]$
- ▶ **Delegation:** $T_1 \otimes T_2 = \min(T_1, T_2) \cdot \delta$ (sequential)
- ▶ **Aggregation:** $T_1 \oplus T_2 = \max(T_1, T_2)$ (parallel)
- ▶ **Zero:** 0 (complete distrust)
- ▶ **Unit:** 1 (complete trust)

The following theorem is the central security guarantee of the trust calculus: it establishes that trust cannot be “laundered” through delegation chains. No matter how an adversary routes content through trusted intermediaries, each hop reduces effective trust by factor δ .

Theorem (Trust Boundedness)

For any delegation chain of depth d :

$$\mathcal{T}_{i \rightarrow k}^{del} \leq \delta^d \tag{15}$$

Proof.

By induction on d . **Base:** $d = 0 \Rightarrow \mathcal{T} \leq 1$. **Step:**

$$\mathcal{T}^{d+1} = \min(\cdot) \cdot \delta \leq \delta^d \cdot \delta = \delta^{d+1}.$$



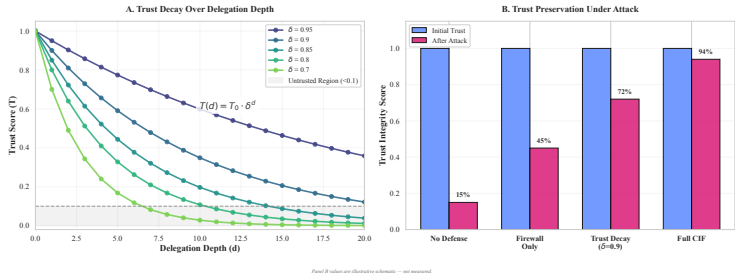
Corollary

Trust cannot be amplified through delegation chains.

Corollary

Trust vanishes exponentially: $\lim_{d \rightarrow \infty} \mathcal{T}_{i \rightarrow k}^{del} = 0$.

Trust Algebra I



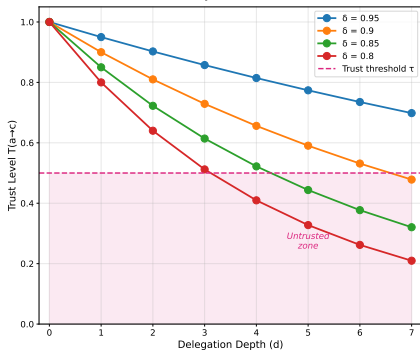
Panel B values are illustrative schematic — not measured.

Figure 2: Trust Decay Over Delegation Depth: Exponential decay curves showing trust attenuation

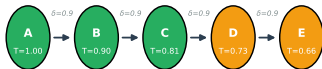
$\mathcal{T}_{\text{del}}^{(d)} = \delta^d \cdot \mathcal{T}_{i \rightarrow j}$ for decay factors $\delta \in \{0.5, 0.7, 0.8, 0.9\}$ over delegation chains of depth $d = 1$ to 10. At $\delta = 0.8$, trust falls below $\tau = 0.5$ at depth 4 and below $\tau = 0.1$ at depth 11 (outside this plotted range). *Note: Values are illustrative examples demonstrating the mathematical framework; specific decay factors should be tuned to deployment context.*

`fig:trust-decay` visualizes the exponential decay of trust across delegation depth for various decay factors δ , demonstrating how the bounded delegation mechanism (`thm:trust-bounded`) prevents trust amplification.

Trust Algebra I

A. Trust Decay: $T(a \rightarrow c) \leq \delta^d \cdot T(a \rightarrow b)$ 

C. Bounded Delegation Chain (Theorem 3.1)



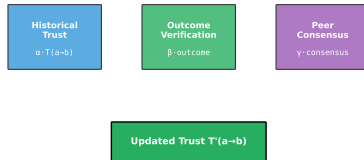
$$T(A \rightarrow E) \leq \delta^4 \cdot T(A \rightarrow B) = 0.9^4 \times 1.0 = 0.66$$

Trust bounded exponentially: prevents trust amplification

B. Trust Update Mechanism

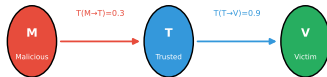
$$T'(a \rightarrow b) = \alpha \cdot T(a \rightarrow b) + \beta \cdot \text{outcome} + \gamma \cdot \text{consensus}$$

where: $\alpha + \beta + \gamma = 1$ (normalization)



D. Trust Laundering Prevention

Attack Attempt: Malicious → Trusted → Target



$$T(M \rightarrow V) \leq \delta \cdot T(M \rightarrow T) = 0.9 \times 0.3 = 0.27 < \tau$$

Delegated trust BLOCKED - below threshold

fig:trust-calculus-comprehensive presents the complete trust calculus mechanics across four panels. Panel A demonstrates the trust decay function $\mathcal{T}(a \rightarrow c) \leq \delta^d \cdot \mathcal{T}(a \rightarrow b)$ for decay factors $\delta \in \{0.8, 0.85, 0.9, 0.95\}$, showing how trust falls below threshold $\tau = 0.5$ at different delegation depths. Panel B formalizes the trust update mechanism $\mathcal{T}'(a \rightarrow b) = \alpha \cdot \mathcal{T}(a \rightarrow b) + \beta \cdot \text{outcome} + \gamma \cdot \text{consensus}$ where $\alpha + \beta + \gamma = 1$, integrating historical trust, outcome verification, and peer consensus. Panel C illustrates a bounded delegation chain (Theorem~11): starting from $\mathcal{T}(A \rightarrow B) = 1.0$ with $\delta = 0.9$, trust decays through agents B, C, D to E with $\mathcal{T}(A \rightarrow E) = 0.9^4 \times 1.0 = 0.66$. Panel D demonstrates

trust laundering prevention: a malicious agent M with $\mathcal{T}(M \rightarrow T) = 0.3$ attempting to exploit trusted intermediary T with $\mathcal{T}(T \rightarrow V) = 0.9$ cannot achieve sufficient delegated trust since $\mathcal{T}(M \rightarrow V) \leq 0.9 \times 0.3 = 0.27 < \tau$, blocking the attack.

Theorem (Delegation Associativity)

The sequential delegation operation is defined by $T_1 \otimes T_2 = \delta \min(T_1, T_2)$. Because the decay factor is applied at each composition step, this operation is not associative in general; chain evaluation therefore preserves the explicit left-to-right delegation order.

$$T_1 \otimes (T_2 \otimes T_3) \neq (T_1 \otimes T_2) \otimes T_3 \quad (16)$$

Proof.

For example, with $(T_1, T_2, T_3, \delta) = (0.9, 0.9, 0.1, 0.9)$, the left-associated value is 0.090 while the right-associated value is 0.081. Thus any associative abstraction must be defined separately (for example, as a chain functional $\min_i T_i \delta^d$), rather than inferred for $\otimes$. □

Theorem (Aggregation Properties)

Trust aggregation $\oplus$ satisfies: (i) associativity, (ii) commutativity, (iii) idempotence, (iv) identity $T \oplus 0 = T$, and (v) absorption $T \oplus 1 = 1$.

Theorem (No Trust Amplification)

For any path $p = (a_0, \dots, a_k)$:

$$\mathcal{T}_{a_0 \rightarrow a_k}^{path} \leq \min_{i \in [0, k-1]} \mathcal{T}_{a_i \rightarrow a_{i+1}} \quad (17)$$

A proof of the no-amplification claim is given in the supplementary material (thm:trust-amp-restated, S01), which states the identical bound under a dedicated label.

Theorem (Trust Monotonicity)

Delegation is monotonic:

$$T_1 \leq T'_1 \wedge T_2 \leq T'_2 \Rightarrow T_1 \otimes T_2 \leq T'_1 \otimes T'_2.$$

When agents operate across modalities—processing text, code, images, audio, and structured data—trust must account for modality-specific reliability and attack susceptibility.

Definition (Modality Trust Adjustment)

For agent a_j operating in modality m , the adjusted trust from agent a_i is:

$$\mathcal{T}_{i \rightarrow j}^m = \mathcal{T}_{i \rightarrow j} \cdot \eta_m \quad (18)$$

where $\eta_m \in (0, 1]$ is the modality reliability factor.

Table 4: Recommended modality reliability factors.

Modality m	η_m	Rationale
Text (verified source)	1.0	Baseline modality
Code (compilable)	0.95	Syntax verification possible
Structured data (schema-valid)	0.90	Schema provides partial verification
Text (external)	0.80	Injection risk
Images	0.70	Adversarial perturbation vulnerability
Audio	0.65	Ultrasonic injection, splicing attacks
Video	0.60	Combines image and temporal vulnerabilities

Theorem (Cross-Modality Delegation Bound)

For delegation chain crossing modalities $m_1, \dots, m_k$:

$$\mathcal{T}_{i \rightarrow j}^{\text{cross}} \leq \delta^d \cdot \prod_{l=1}^k \eta_{m_l} \quad (19)$$

This ensures that trust degradation compounds across both delegation depth and modality transitions, preventing adversaries from laundering low-trust content through modality boundaries.

In enterprise deployments, multiagent systems increasingly span organizational boundaries. A financial services orchestrator might delegate to a risk assessment system from one vendor, a compliance checker from another, and market data feeds from multiple providers. **Federated trust** addresses how to reason about trust across these boundaries.

Definition (Trust Domain)

A trust domain $\mathcal{D}$ is a set of agents sharing a common trust authority and consistent trust semantics.

Definition (Cross-Domain Trust)

For agent a_i in domain $\mathcal{D}_1$ and agent a_j in domain $\mathcal{D}_2$:

$$\mathcal{T}_{i \rightarrow j}^{\text{fed}} = \mathcal{T}_{i \rightarrow \mathcal{D}_2} \cdot \mathcal{T}_{\mathcal{D}_2}(j) \quad (20)$$

where $\mathcal{T}_{i \rightarrow \mathcal{D}_2}$ is a_i 's trust in domain $\mathcal{D}_2$ and $\mathcal{T}_{\mathcal{D}_2}(j)$ is a_j 's standing within its domain.

This two-stage model captures realistic trust reasoning: an organization might trust a vendor (domain trust) differently than individual agents within that vendor (agent trust).

Property (Federated Trust Bound)

Cross-domain trust is bounded by domain trust:

$$\mathcal{T}_{i \rightarrow j}^{fed} \leq \mathcal{T}_{i \rightarrow \mathcal{D}_2} \quad (21)$$

ensuring that untrusted domains cannot boost individual agent trust.

Federated trust introduces additional challenges that remain open research problems:

- ▶ **Trust semantics heterogeneity:** Different domains may use incompatible trust scales or update rules
- ▶ **Trust attestation:** How can domains cryptographically attest to their internal trust assessments?
- ▶ **Privacy-preserving trust:** Can trust be verified without revealing sensitive internal assessments?

Definition (Evidence Structure)

Evidence is a tuple $e = \langle \phi, c, s, \pi \rangle$ comprising proposition ϕ , confidence $c \in [0, 1]$, source s , and provenance chain π .

Definition (Trust-Weighted Bayesian Update)

Upon receiving evidence e from source s :

$$\mathcal{B}_i^{t+1}(\phi) = \frac{\mathcal{B}_i^t(\phi) \cdot [\mathcal{T}_{i \rightarrow s} P(e|\phi) + (1 - \mathcal{T}_{i \rightarrow s}) P_0(e|\phi)]}{\sum_{\psi} \mathcal{B}_i^t(\psi) \cdot [\mathcal{T}_{i \rightarrow s} P(e|\psi) + (1 - \mathcal{T}_{i \rightarrow s}) P_0(e|\psi)]} \quad (22)$$

Trust mixes the source likelihood with a neutral likelihood P_0 ; low-trust sources therefore have a diminished update impact rather than canceling as a common normalization factor.

Rule B-Direct (Direct Evidence):

$$\frac{e = \langle \phi, c, s, \pi \rangle \quad V(\pi) = 1 \quad \mathcal{T}_{i \rightarrow s} \geq \tau_{\text{trust}}}{\mathcal{B}_i^{t+1}(\phi) = \text{BayesUpdate}(\mathcal{B}_i^t(\phi), c \cdot \mathcal{T}_{i \rightarrow s})} \quad (23)$$

Rule B-Delegated (Delegated Evidence): evidence received through a delegation chain of depth d is accepted only when its provenance is valid and its delegated trust meets the configured threshold:

$$\frac{e = \langle \phi, c, s, \pi \rangle \quad V(\pi) = 1 \quad \mathcal{T}_{i \rightarrow s}^{\text{del}} = \mathcal{T}_{i \rightarrow s}^{(d)} \geq \tau_{\text{trust}}}{\mathcal{B}_i^{t+1}(\phi) = \text{BayesUpdate}(\mathcal{B}_i^t(\phi), c \cdot \mathcal{T}_{i \rightarrow s}^{\text{del}})} \quad (24)$$

Here $\mathcal{T}_{i \rightarrow s}^{(d)} \leq \delta^d$ is the delegated trust after d hops, making depth-dependent attenuation explicit.

Rule B-Corroboation (Multiple Sources):

$$\frac{\{e_j\}_{j=1}^k : \forall j. e_j = \langle \phi, c_j, s_j, \pi_j \rangle \quad |\{s_j\}| \geq \kappa}{\mathcal{B}_i^{t+1}(\phi) = 1 - \prod_{j=1}^k (1 - c_j \cdot \mathcal{T}_{i \rightarrow s_j})} \quad (25)$$

Lemma (Belief Boundedness)

After any update sequence: $\forall \phi : 0 \leq \mathcal{B}_i(\phi) \leq 1$.

Definition (Sandboxed Beliefs)

Beliefs from unverified sources enter provisional state:

$$\mathcal{B}_i = \mathcal{B}_{\text{verified}} \cup \mathcal{B}_{\text{provisional}} \quad (26)$$

Rule S-Sandbox (Enter Sandbox):

$$\frac{e = \langle \phi, c, s, \pi \rangle \quad (\mathcal{T}_{i \rightarrow s} < \tau_{\text{trust}} \vee V(\pi) = 0)}{\mathcal{B}_{\text{prov}} \leftarrow \mathcal{B}_{\text{prov}} \cup \{(\phi, c, s, \pi, \text{TTL})\}} \quad (27)$$

Rule S-Promote (Sandbox Promotion):

$$\frac{(\phi, \dots) \in \mathcal{B}_{\text{prov}} \quad V(\pi) = 1 \quad \text{Consistent}(\mathcal{B}_{\text{ver}} \cup \{\phi\}) \quad |\text{Corr}(\phi)| \geq \kappa}{\mathcal{B}_{\text{ver}} \leftarrow \mathcal{B}_{\text{ver}} \cup \{\phi\}; \quad \mathcal{B}_{\text{prov}} \leftarrow \mathcal{B}_{\text{prov}} \setminus \{(\phi, \dots)\}} \quad (28)$$

Rule S-Expire (Sandbox Expiry):

$$\frac{(\phi, c, s, \pi, \text{TTL}) \in \mathcal{B}_{\text{prov}} \quad \text{TTL} \leq 0}{\mathcal{B}_{\text{prov}} \leftarrow \mathcal{B}_{\text{prov}} \setminus \{(\phi, c, s, \pi, \text{TTL})\}} \quad (29)$$

Promotion requires: (1) provenance verification $V(\pi) = 1$, (2) consistency with verified beliefs, and (3) corroboration threshold κ .

Information-Theoretic Detection Bounds

Having established the trust calculus (how agents reason about each other) and belief update semantics (how agents incorporate information), we now turn to fundamental limits on attack detection. This section establishes fundamental limits on what any detection system can achieve. Like Shannon's channel capacity in communications, these bounds are not limitations of specific mechanisms but mathematical constraints on what is possible.

Definition (Attack Information Channel)

An attack models a communication channel from adversary to target:

$$\text{Channel} : \mathcal{A}_{\text{adv}} \rightarrow \sigma_{\text{target}} \quad (30)$$

Theorem (Minimum Attack Entropy)

For attack $\mathcal{A}$ to succeed with probability p :

$$H(\mathcal{A}) \geq -\log_2(1-p) + H(\sigma_{\text{target}}|\mathcal{A}) \quad (31)$$

Proof.

By the data processing inequality. The attack must contain sufficient information to change the target state. □

Corollary

Attacks with low entropy (simple patterns) are more detectable.

Definition (Detector Information Gain)

For detector D observing system state S :

$$I(D; \mathcal{A}) = H(\mathcal{A}) - H(\mathcal{A}|D(S)) \quad (32)$$

Theorem (Fundamental Detection Limit)

No detector achieves detection rate r if:

$$r > \frac{I(D; \mathcal{A})}{H(\mathcal{A})} \quad (33)$$

The following theorem captures the fundamental tradeoff facing attackers: high-impact attacks are easier to detect, while stealthy attacks have limited effect. This is not a limitation of our defenses—it is a mathematical constraint that any attack must satisfy.

Theorem (Stealth-Impact Tradeoff)

For attack with impact $\mathcal{I}$ and stealth $\mathcal{S}$ (inverse detectability):

$$\mathcal{I} \cdot \mathcal{S} \leq C_{\text{channel}} \quad (34)$$

where C_{channel} is the attack channel capacity.

Information-Theoretic Detection Bounds I

Remark (Geometric Sharpening of the Stealth-Impact Bound)

Theorem 31 can be sharpened using the information geometry of the belief manifold (Part 2, Supplement S10). When belief space is endowed with the Fisher-Rao metric $G_{ij}(p) = \delta_{ij}/p_i$ [?], the Fisher-Rao geodesic distance

$$d_{\text{FR}}(p, q) = 2 \arccos \left(\sum_i \sqrt{p_i q_i} \right)$$

measures the geometric “distance” of an attack. Impact $\mathcal{I}$ is proportional to $d_{\text{FR}}(p_0, p_{\text{attacked}})$, while stealth $\mathcal{S}$ is inversely proportional to the same quantity. The curvature of the probability simplex Δ^{n-1} (constant positive curvature $\kappa = n(n-1)/4$) implies that the product $\mathcal{I} \cdot \mathcal{S}$ is bounded by the diameter of the manifold:

$$\mathcal{I} \cdot \mathcal{S} \leq \frac{\pi}{2},$$

since $d_{\text{FR}} \leq \pi$ (Hellinger antipodal distributions). This recovers $C_{\text{channel}} = \pi/2$ as the tight bound when impact is measured in natural (Fisher-Rao) units. It is a geometric argument for the constant in Theorem 31, not a proof of that theorem: the theorem itself is stated without proof and is recorded as deferred in the

Information-Theoretic Detection Bounds I

Table 5: Information-theoretic bounds by attack type.

Attack Type	Min Entropy $H(\mathcal{A})$	Detection Lower Bound
Belief Injection	$\log_2 \Phi $	$\frac{\log_2 \Phi }{H(\mathcal{B})}$
Goal Hijacking	$H(\mathcal{G}_{\text{target}})$	$\frac{H(\mathcal{G}_{\text{target}})}{H(\mathcal{G})}$
Trust Manipulation	$\log_2 n$	$\frac{\log_2 n}{H(\mathcal{T})}$
Progressive Drift	$T \cdot \log_2(1/\delta_{\text{step}})$	$\frac{T}{w} \cdot \delta_{\text{step}}$

Theorem (Progressive Attack Detection Bound)

For progressive drift attack with step size δ over T steps:

$$P(\text{detect within } w \text{ steps}) \leq 1 - \left(1 - \frac{\delta}{\theta_{\text{step}}}\right)^w \quad (35)$$

Corollary

Smaller step sizes exponentially reduce detection probability but require more time for impact.

Definition (Defense Information Budget)

Total monitoring capacity: $B_{\text{total}} = \sum_{d \in \mathcal{D}} B_d$.

Theorem (Optimal Budget Allocation)

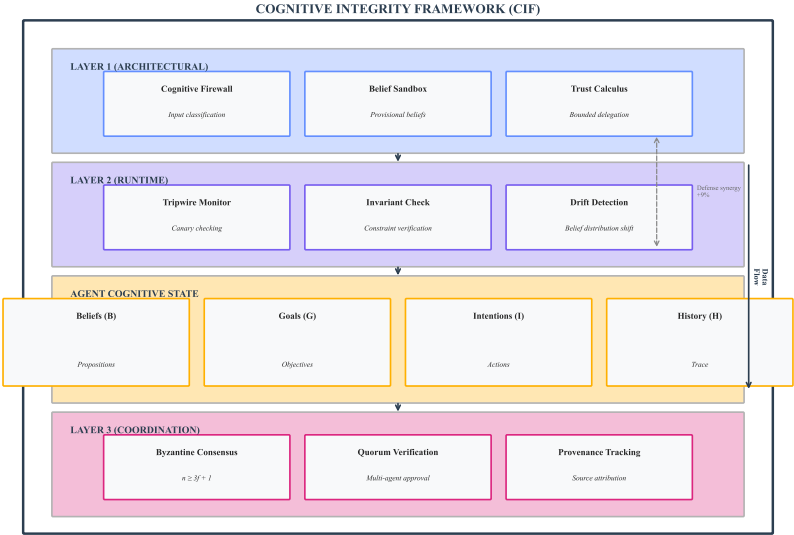
Given attack distribution $P(\mathcal{A}_k)$ and detector sensitivities $\eta_d(\mathcal{A}_k)$:

$$B_d^* = \frac{\sum_k P(\mathcal{A}_k) \cdot \eta_d(\mathcal{A}_k)}{\sum_{d'} \sum_k P(\mathcal{A}_k) \cdot \eta_{d'}(\mathcal{A}_k)} \cdot B_{total} \quad (36)$$

Proof.

Lagrangian optimization maximizing expected detection subject to budget constraint. □

Information-Theoretic Detection Bounds I



Information-Theoretic Detection Bounds II

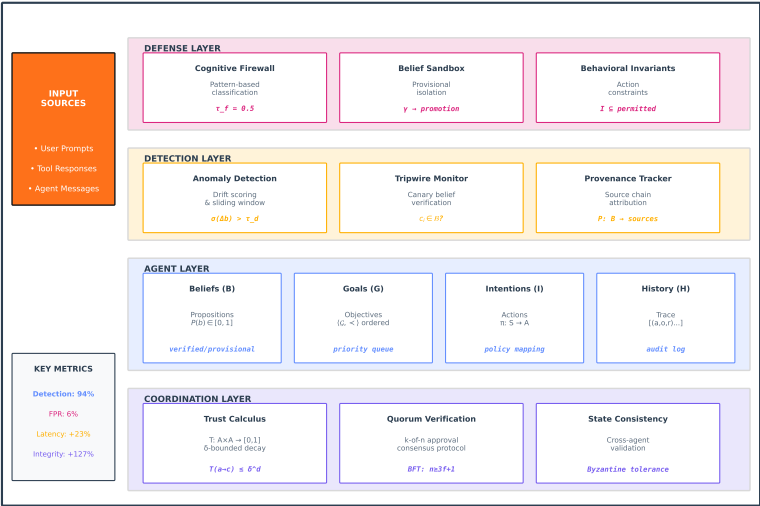
Figure 4: CIF Architecture Overview: Three-layer defense architecture—**Layer 1** (Architectural): Cognitive Firewall at entry, Belief Sandbox for unverified data, Trust Calculus for delegation; **Layer 2** (Runtime): Tripwires for belief monitoring, Invariant verification, Drift detection; **Layer 3** (Coordination): Byzantine consensus, Quorum verification, Provenance tracking.

`fig:cif-architecture` presents the layered CIF architecture with architectural defenses (left), runtime defenses (center), and coordination mechanism (right). `fig:cif-comprehensive` expands this to show data flow, attack interception points, and defense composition.

Information-Theoretic Detection Bounds I

COGNITIVE INTEGRITY FRAMEWORK (CIF)

Layered Defense Architecture for Multiagent AI Systems



Information-Theoretic Detection Bounds II

Figure 5: Comprehensive CIF Architecture: Extended architecture showing data flow from user input through all defense layers to agent output. Attack interception points labeled Ω_1 – Ω_5 indicate where each adversary class is detected. Defense composition follows multiplicative detection rate improvement (`cor:layered-defense`).

`fig:cif-comprehensive` provides a detailed view of the complete CIF architecture, including all component formulas and their interactions. The defense layer implements the cognitive firewall with threshold $\tau_f = 0.5$, the belief sandbox with promotion function γ , and behavioral invariants constraining intentions $\mathcal{I} \subseteq$ permitted. The detection layer specifies anomaly scoring $\sigma(\Delta b) > \tau_d$, tripwire verification $c_i \in \mathcal{B}?$, and provenance tracking $P : \mathcal{B} \rightarrow$ sources. The coordination layer encodes the trust calculus $\mathcal{T} : \mathcal{A} \times \mathcal{A} \rightarrow [0, 1]$ with δ -bounded decay, k-of-n quorum protocols, and Byzantine fault tolerance ($n \geq 3f + 1$). For empirical validation of detection rates and performance overhead, see Part 2 of this series.

CIF-AD Integration: Action-Delegation Coupling

CIF-AD Integration: Action-Delegation Coupling I

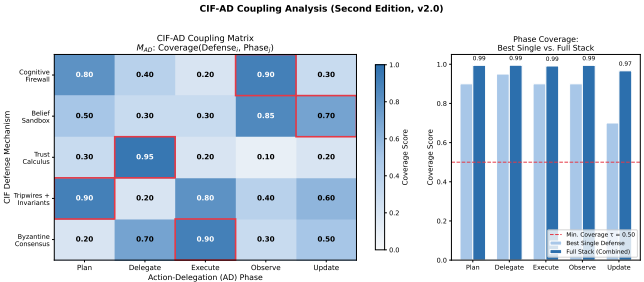


Figure 6: CIF-AD action–delegation coupling matrix showing how Functional Requirements map to Design Parameters and delegation actions, making explicit where a Goal-Hijacking transient can introduce off-diagonal coupling.

The Cognitive Integrity Framework provides static structural guarantees (trust bounds, belief consistency). But multiagent operators are dynamic: agents act, delegate, and observe outcomes. This section formalizes how CIF integrates with the Action-Delegation (AD) cycle to provide continuous runtime security guarantees.

The Action-Delegation Cycle I

Definition (Action-Delegation Cycle)

The Action-Delegation (AD) cycle for agent a_i at time t is:

$$AD_i^t : \sigma_i^t \xrightarrow{\text{plan}} \mathcal{J}_i^t \xrightarrow{\text{delegate}} \langle a_j, \text{task}_j \rangle \xrightarrow{\text{execute}} \mathcal{H}_i^{t+1} \xrightarrow{\text{update}} \sigma_i^{t+1} \quad (37)$$

comprising: cognitive state assessment, intention formation, delegation decision, execution observation, and state update.

The Action-Delegation Cycle I

Each phase of the AD cycle introduces distinct attack surfaces that CIF must protect:

The Action-Delegation Cycle I

Table 6: AD cycle phases and corresponding CIF defenses.

AD Phase	Attack Surface	CIF Mechanism	Formal Guarantee
Plan	Goal injection	Invariant enforcement	$\mathcal{G}_i \subseteq \mathcal{G}_{\text{principal}}$
Delegate	Trust exploitation	Trust calculus	$\mathcal{T}^{\text{del}} \leq \delta^d$
Execute	Result tampering	Provenance tracking	$V(\pi(\phi)) = 1$
Observe	Evidence fabrication	Belief sandbox	$\mathcal{B}_{\text{prov}}$ isolation
Update	Drift induction	KL drift detection	$D_{\text{KL}} < \theta_{\text{drift}}$

Definition (CIF-AD Coupling Matrix)

The coupling matrix $\mathbf{M}_{AD} \in \mathbb{R}^{5 \times 5}$ quantifies the degree to which each CIF defense addresses each AD phase:

$$[\mathbf{M}_{AD}]_{ij} = \text{Coverage}(\text{Defense}_i, \text{Phase}_j) \in [0, 1] \quad (38)$$

CIF-AD Coupling Matrix I

Table 7: CIF-AD coupling matrix: coverage of each defense mechanism across AD cycle phases.

Defense	Plan	Delegate	Execute	Observe	Update
Cognitive Firewall	0.80	0.40	0.20	0.90	0.30
Belief Sandbox	0.50	0.30	0.30	0.85	0.70
Trust Calculus	0.30	0.95	0.20	0.10	0.20
Tripwires + Invariants	0.90	0.20	0.80	0.40	0.60
Byzantine Consensus	0.20	0.70	0.90	0.30	0.50

Theorem (CIF-AD Full Coverage)

The CIF defense stack provides non-zero coverage for every AD cycle phase:

$$\forall j \in \{1, \dots, 5\} : \max_i [\mathbf{M}_{AD}]_{ij} > 0 \quad (39)$$

Moreover, the minimum column coverage exceeds $\tau_{\text{cov}} = 0.50$ for every phase.

Proof.

By inspection of Table 7: minimum column maxima are Plan (0.90), Delegate (0.95), Execute (0.90), Observe (0.90), Update (0.70). All exceed $\tau_{\text{cov}} = 0.50$. □

CIF-OODA Integration: Temporal Security Across Decision Cycles

CIF-OODA Integration: Temporal Security Across Decision Cycles I

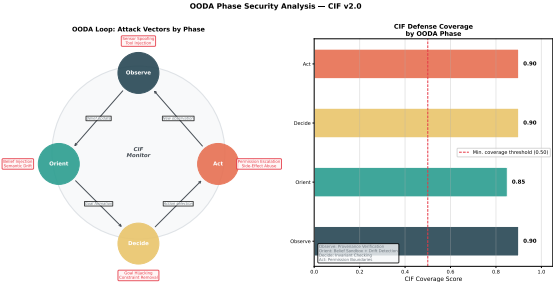


Figure 7: CIF-OODA phase diagram mapping the defense mechanisms onto the Observe–Orient–Decide–Act cycle, indicating at which phase each defense intervenes and its latency budget.

The OODA loop (Observe-Orient-Decide-Act) models how agents process information and take action over time. Cognitive attacks often target specific OODA phases to corrupt decision-making while minimizing detectable footprint. This section formalizes the OODA-phase attack model and CIF's phase-specific defenses.

Definition (OODA State Machine)

Agent a_i 's OODA loop is a state machine:

$$\text{OODA}_i = (Q, q_0, \Sigma, \delta_{\text{OODA}}) \quad (40)$$

where $Q = \{\text{Observe, Orient, Decide, Act}\}$ are states, $q_0 = \text{Observe}$ is the initial state, $\Sigma = \mathcal{M} \cup \text{Events}$ is the input alphabet, and $\delta_{\text{OODA}} : Q \times \Sigma \rightarrow Q$ is the transition function.

The OODA loop maps to cognitive state components:

Observe $\leftrightarrow \mathcal{H}_i$ (history accumulation) (41)

Orient $\leftrightarrow \mathcal{B}_i$ (belief update) (42)

Decide $\leftrightarrow \mathcal{G}_i, \mathcal{I}_i$ (goal selection, intention formation) (43)

Act $\leftrightarrow \text{Actions}(a_i)$ (execution) (44)

Definition (OODA Phase Attack)

An attack $\mathcal{A}_{\text{OODA}}^{(q)}$ targets OODA phase $q \in Q$:

$$\mathcal{A}_{\text{OODA}}^{(q)} : \text{Corrupt}(\sigma_i \mid q) \text{ such that } \text{OODA}_i \text{ produces adversarial outcome at Act} \quad (45)$$

Phase-Specific Attack Taxonomy I

Table 8: OODA phase attacks and CIF defenses.

OODA Phase Attack Vector		CIF Defense	Detection Mechanism
Observe	Sensor spoofing, tool injection	Provenance verification	Source taint labels
Orient	Belief injection, semantic drift	Belief sandbox, drift detection	KL divergence monitoring
Decide	Goal hijacking, constraint removal	Invariant checking	Goal alignment verification
Act	Permission escalation, side effects	Permission boundaries	Action audit logging

OODA Temporal Coupling I

Definition (OODA Cycle Time)

The OODA cycle time τ_{OODA} is the duration from Observe initiation to Act completion. The CIF monitoring overhead must satisfy:

$$\tau_{\text{CIF}} \leq \epsilon_{\text{OODA}} \cdot \tau_{\text{OODA}} \quad (46)$$

where $\epsilon_{\text{OODA}} \in (0, 0.1)$ is the maximum tolerable overhead fraction.

Property (OODA-CIF Latency Compatibility)

The CIF defense stack (Table 23) with total latency $L_{\text{total}} \leq 141\text{ms}$ satisfies the overhead constraint for OODA cycles $\tau_{\text{OODA}} \geq 1.41\text{s}$ (human-scale operations). For faster OODA cycles (drone swarms: $\tau \approx 100\text{ms}$), subset selection is required:

$$\mathcal{D}^* = \arg \max_{\mathcal{D}' \subseteq \mathcal{D}} P_{\text{detect}}(\mathcal{D}') \quad \text{s.t.} \quad L(\mathcal{D}') \leq \epsilon_{\text{OODA}} \cdot \tau_{\text{OODA}} \quad (47)$$

Theorem (CIF-OODA Security Invariant)

Under CIF-OODA integration, an agent completes each OODA cycle with cognitive integrity preserved with probability:

$$P(\text{Integrity}(a_i^{t+1}) \mid \text{Integrity}(a_i^t)) \geq 1 - \prod_{d \in \mathcal{D}} (1 - P_d^{\text{OODA}}) \quad (48)$$

where P_d^{OODA} is the detection probability of defense d across all four OODA phases.

Proof.

Integrity is violated only if an attack passes all defenses in all OODA phases it targets. By defense independence (Lemma 11.7), the probability of passing all defenses is $\prod_d (1 - P_d^{\text{OODA}})$. Integrity is preserved with complementary probability. $\square$